

CHAPTER 1

INTRODUCTION

- Normally exhaust fans are installed in kitchens, including restaurant kitchens, which keep running continuously whether smoke is present or not. Here is a small project that uses electronic devices that switches on the exhaust fan automatically only when there is smoke in the air. It switches the exhaust fan off when the smoke has been removed.
- There by the Electricity or the power consumption is reduced.
- Hence the efficiency of the device is increased.
- Due to continuous usage of exhaust fan the major issue is the noise produced which can be reduced by the use of this project
- The beauty of this system lies in its responsiveness, ensuring that fresh air circulates effortlessly, creating a healthier atmosphere without the need for manual intervention. The harmony between technology and the natural flow of air offers an elegant solution to indoor air quality challenges, making this automatic exhaust fan not just a functional device, but a smart companion that adapts to the needs of modern living. With this innovation, the air you breathe becomes cleaner, safer, and more controlled—an essential feature for homes, offices, and any space prioritizing health and well-being.

1.1 OBJECTIVES

- Automatic Operation: Responds to smoke detection without manual intervention, offering convenience and timely response to potential hazards.
- Energy Efficiency: Operates only when smoke is detected. Conserving energy compared to continuous running fans.
- Minimized Smoke Damage: Reduces smoke accumulation, potentially preventing damage to equipment, furniture, and structures.
- Fire Prevention and Safety: Automatically activates to vent smoke, reducing the risk of fire hazards and ensuring safety in environments prone to smoke accumulation.

- **Health Protection:** Reduces exposure to smoke and harmful particles protecting the health of occupants in space.

1.2 AIM OF THE PROJECT

Improved Air Quality: To ensure the continuous circulation of fresh air by removing harmful gases like carbon dioxide and volatile organic compounds (VOCs), thereby maintaining a healthy environment indoors.

User Convenience: To create a hassle-free experience where users can rely on the system to maintain optimal air conditions without constant monitoring or manual adjustments

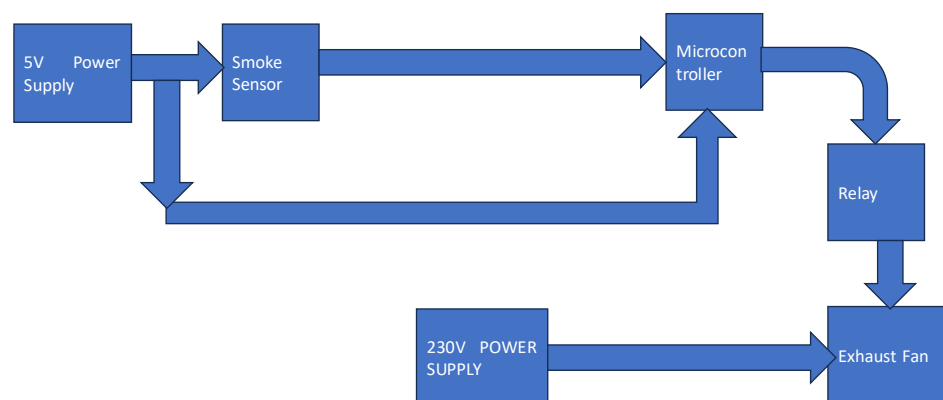
CHAPTER 2

PROPOSED METHODOLOGY

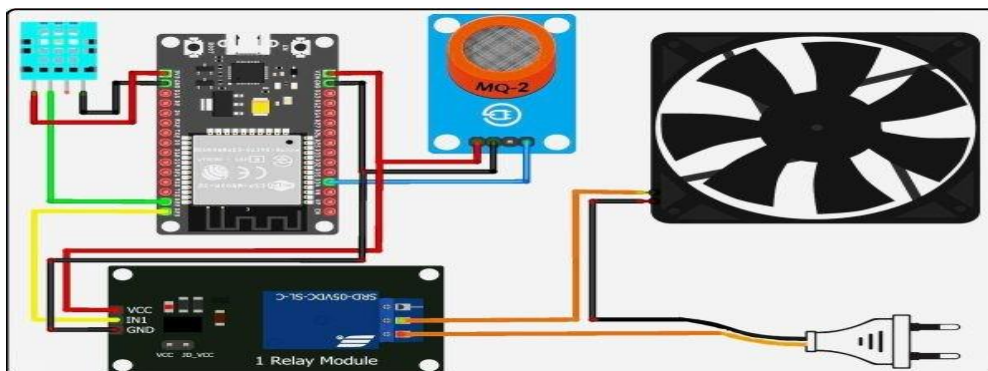
The proposed system consists of two main parts: Hardware and Software. The hardware consists of a Gas sensor, 5v 1 channel relay, buzzer, led's and an ESP8266 node MCU Board. Software includes an Arduino Software (IDE).

The Gas sensor is connected to the node MCU board it senses the smoke and responds with respect to the threshold given. The node MCU board is provided a supply voltage of 5v. The relay is also connected to the 3.3v pin of the node MCU board which provides input to the relay. The led's and relay are connected to the digital pins of the ESP8266. When the gas percentage is less than the threshold mentioned green led is turned on indicating that no smoke is present. When the gas percentage is Greater than the threshold mentioned red led is turned on indicating that smoke is present. When smoke is present buzzer is turned on and relay is also turned on. The relay controls the on and off operation of an exhaust fan.

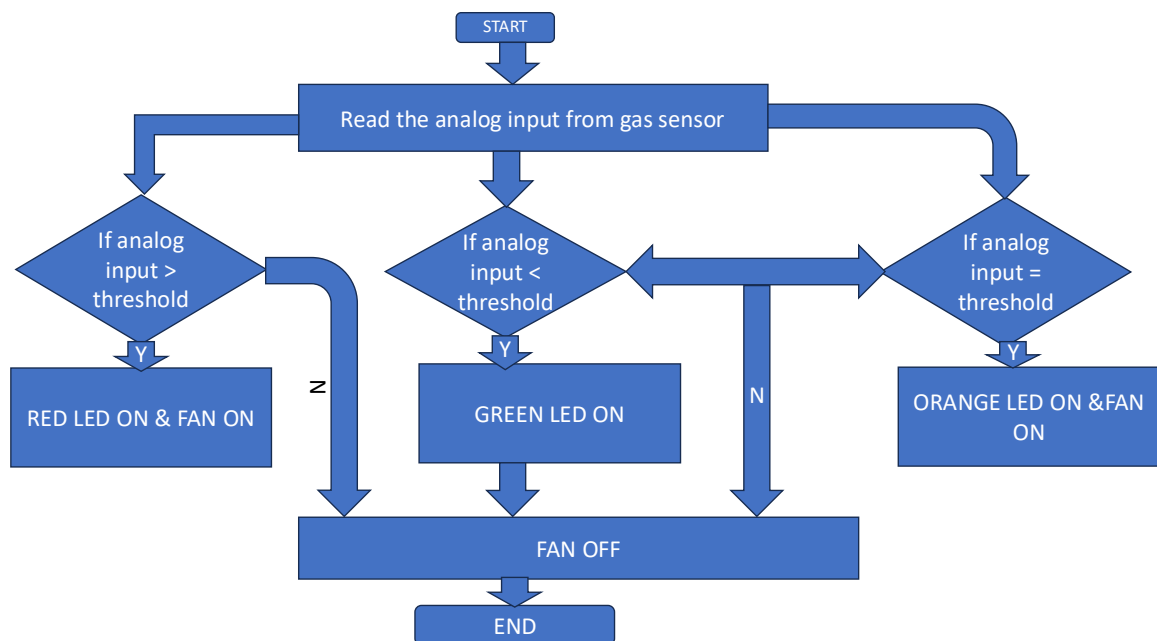
2.1 BLOCK DIAGRAM



2.2 SCHEMATIC DIAGRAM



2.3 FLOW CHART



CHAPTER 3

HARDWARE REQUIREMENTS

➤ Hardware Components:

- 1. Smoke Detection Sensors
- MQ-2 Sensor: This is a popular gas sensor capable of detecting smoke, LPG, carbon monoxide, methane, and other gases. It can be used to monitor the presence of smoke in the air and send signals when smoke levels exceed a certain threshold
- 2. ESP8266/ESP32: These are microcontrollers with built-in Wi-Fi capabilities, ideal for integrating IoT-based control and monitoring of the exhaust fan from remote locations.
- 3. Relay Module
- Relay: A relay acts as a switch that is controlled by the microcontroller to turn on or off the exhaust fan. A relay allows the microcontroller to control high-voltage devices like an exhaust fan, which requires more power than the microcontroller can handle directly.
- 4. Power Supply
- DC Power supply of 5volts for the operation microcontroller
- 5. Software / Programming Tools
- Arduino IDE: For programming the Arduino microcontroller. The IDE offers a user-friendly interface to write code and upload it to the microcontroller.
- 6. DC MOTOR Type: Standard 130 Type DC motor, Operating Voltage: 4.5V to 9V, Current at No load: 70mA (max), No-load Speed: 9000rpm
- Loaded current: 250mA (approx)
- Motor Size: 27.5mm x 20mm x 15mm, Weight: 17 grams

Components:

ESP8266 MODULE



1CHANNEL RELAY MODULE



LED'S



MQ2 GAS SENSOR



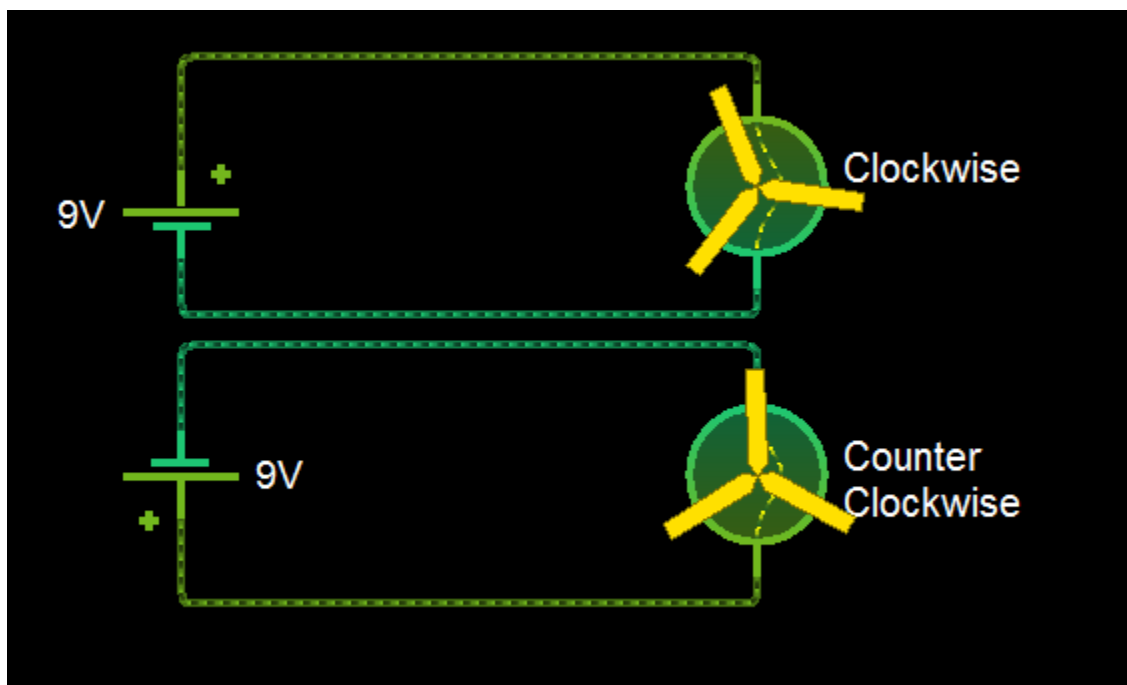
DC MOTOR



BUZZER



Woking of an exhaust fan



CHAPTER 4

SOFTWARE REQUIREMENTS

4.1 Arduino ide software:

The program code written in Arduino IDE is known as a sketch. This IDE contains the following parts in it.

- **Text editor:** This is where the interpreted code can be written using a simplified version of C++ programming language.
- **Message section:** It shows errors and gives feedback on saving and exporting the code.
- **Text:** The soothe displays text output by the Arduino environment along with complete error messages and other information
- **Console Toolbar:** This toolbar encompasses various buttons like Verify, Upload, New, Open, Save and Serial Monitor. On the bottom right-hand corner of the window there displays the Development Board and the Serial Port in use.





Set Up Arduino IDE: Install the Arduino IDE (if not already installed) on your computer to program the microcontroller.

Write the Code:

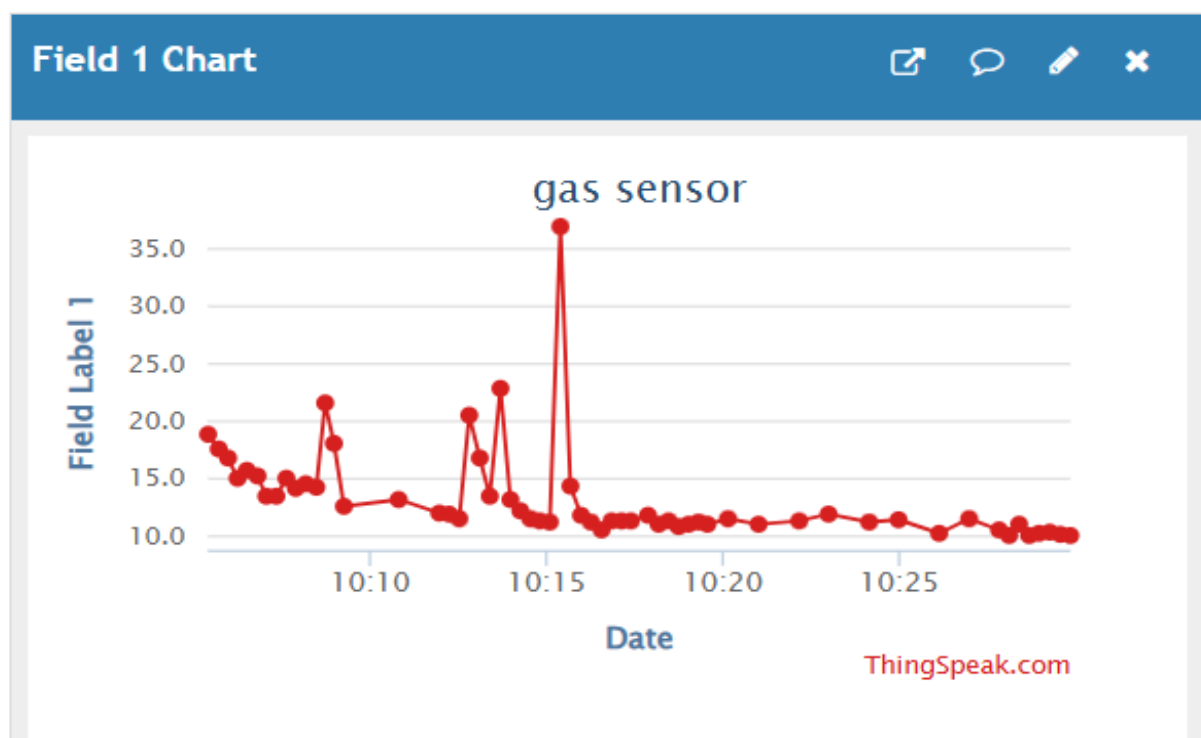
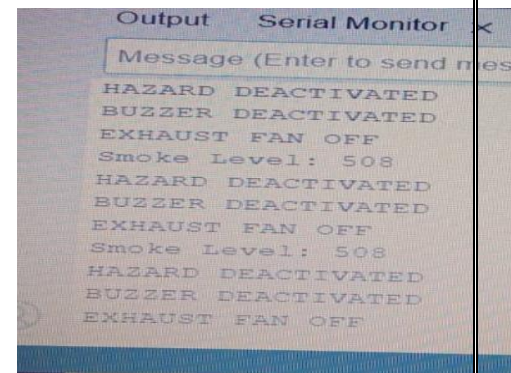
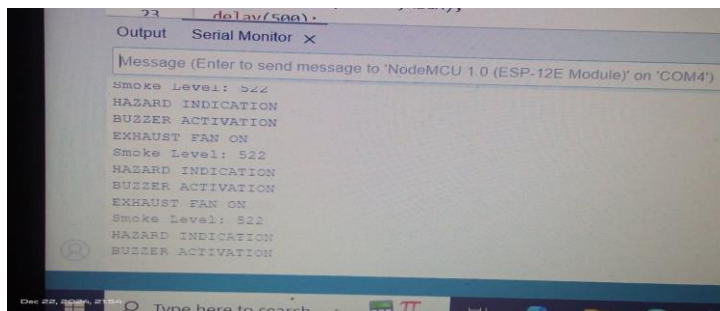
- Initialize the sensor and the relay pin in the setup () function.
- Continuously read the smoke sensor's analog output to check if smoke is detected.
- Define a threshold value of the sensor's output (e.g., if the value is higher than a set threshold, smoke is detected).
- If smoke is detected, activate the relay (turn on the exhaust fan).
- If no smoke is detected, deactivate the relay (turn off the fan).

CHAPTER 5

RESULTS

Experimental result

The project successfully collected smoke level



CHAPTER 6

CONCLUSION AND FUTURE WORK

- **The Automatic Exhaust Fan** project demonstrates an innovative solution for controlling the air circulation in a space based on environmental conditions such as temperature and humidity. The system efficiently manages the exhaust fan operation, reducing energy consumption while ensuring that the environment remains comfortable and well-ventilated.
1. **Automation & Efficiency:** By integrating sensors such as temperature and humidity sensors with a microcontroller, the exhaust fan operates autonomously, turning on or off based on real-time readings. This ensures that the fan only works when necessary, optimizing energy use.
 2. **Improved Air Quality:** The automatic fan helps maintain better indoor air quality by adjusting ventilation as per environmental changes, preventing excess moisture, heat, or odors from accumulating.
 3. **Cost-Effective:** With reduced energy consumption, this system can lead to lower electricity bills. The automatic nature of the system also minimizes human intervention, ensuring that the fan operates at optimal times, further improving overall efficiency.
 4. **User-Friendly:** The system is easy to set up and can be used in various applications, including homes, kitchens, and offices. It provides convenience by eliminating the need for manual control, making it a practical solution for everyday use.

FUTURE SCOPE

Integration with Multiple Sensors

- **Multi-sensor Inputs:** In the future, the system could be upgraded to integrate multiple types of sensors for more comprehensive monitoring:
 - o **Temperature and Humidity Sensors:** For more accurate control, the system could also use temperature and humidity readings to turn the fan on or off. This is particularly useful in preventing mold growth in bathrooms, basements, and kitchens.
 - o **Air Quality Sensors:** Integrating advanced air quality sensors (e.g., CO₂, PM_{2.5}) could allow the system to monitor particulate matter and air pollutants, ensuring better air quality by triggering the fan based on specific thresholds of air contaminants.
 - o **Carbon Monoxide Detection:** In garages, industrial setups, or kitchens, integrating CO sensors could allow the system to activate the fan in response to dangerous levels of carbon monoxide.

Energy Efficiency Improvements

•**Smart Energy Management:** Future versions could include integration with smart grids or energy management systems to optimize energy usage. The system could be designed to monitor energy consumption and reduce fan usage during off-peak hours, or adjust fan speeds for optimal energy efficiency.

•**Fan Speed Control:** Instead of simply turning the fan on and off, implementing variable speed control (through a PWM controller) could allow the fan to operate at a lower speed when the air quality is only slightly poor, conserving more energy.

Noise Level Control: To address the potential issue of fan noise, future iterations could involve integrating noise sensors to monitor the sound levels in the environment.

The system could adjust the fan's speed or activate noise-reducing technologies when the sound exceeds a certain threshold.

- **Quieter Fans:** As part of future improvements, the project could explore quieter exhaust fans or soundproofing methods for the fan housing to ensure that the system's operation is both efficient and quiet.

- **Integration with Building Management Systems (BMS)**

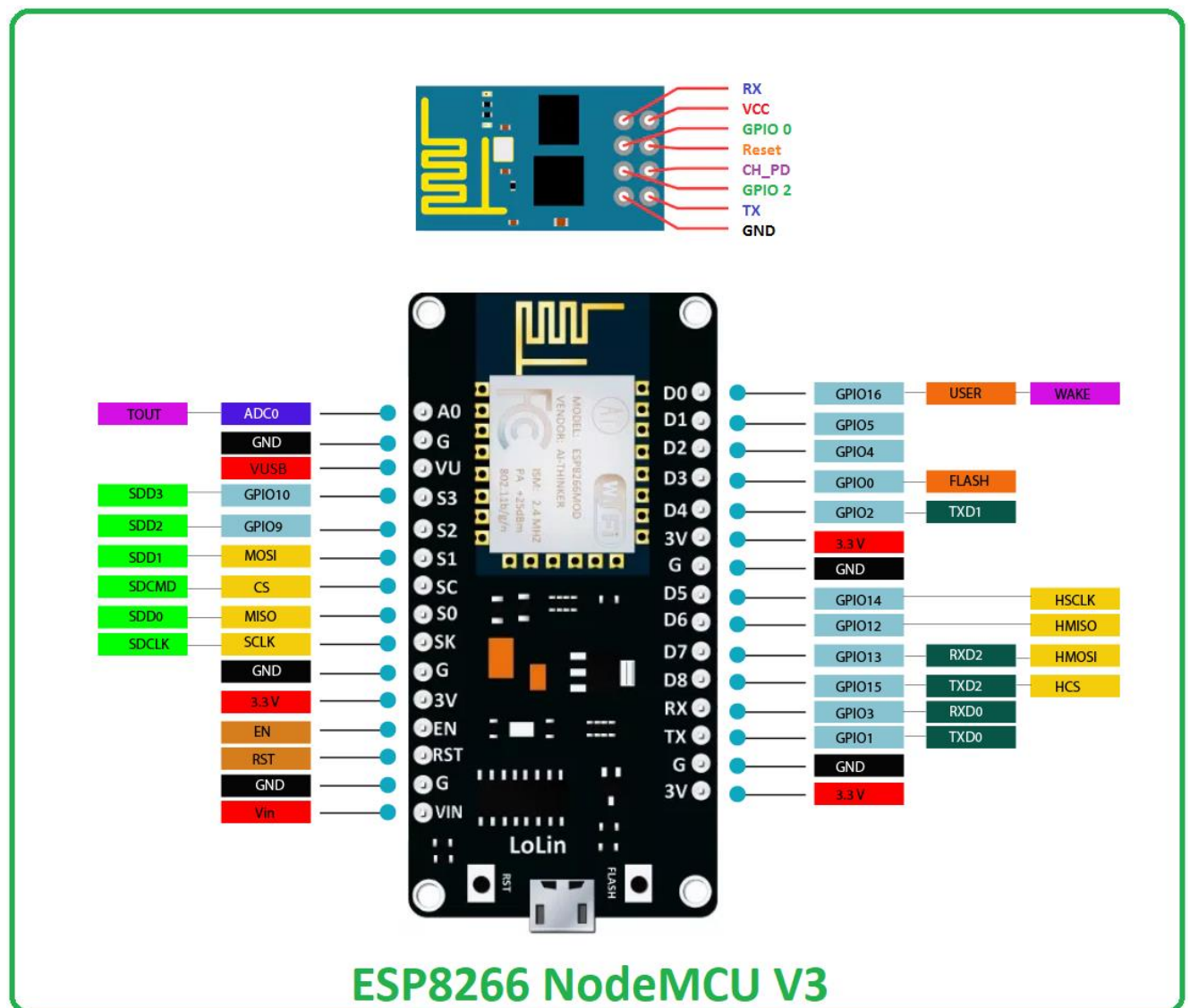
- **Commercial and Large-Scale Applications:** The system could be integrated with **Building Management Systems (BMS)**, which are commonly used in large commercial buildings, hotels, or shopping malls. This would allow centralized control of ventilation, lighting, and other systems to optimize energy use across the building.

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APPENDIX

PIN DIAGRAM OF ESP8266



1	VDDA	P	Analog Power 2.5V ~ 3.6V
2	LNA	I/O	RF antenna interface Chip output impedance= $39+j6\ \Omega$. It is suggested to retain the π -type matching network to match the antenna.
3	VDD3P3	P	Amplifier Power 2.5V ~ 3.6V
4	VDD3P3	P	Amplifier Power 2.5V ~ 3.6V
5	VDD_RTC	P	NC (1.1V)
6	TOUT	I	ADC pin. It can be used to test the power-supply voltage of VDD3P3 (Pin3 and Pin4) and the input power voltage of TOUT (Pin 6). However, these two functions cannot be used simultaneously.
7	CHIP_EN	I	Chip Enable High: On, chip works properly Low: Off, small current consumed
8	XPD_DCDCI/O		Deep-sleep wakeup (need to be connected to EXT_RSTB); GPIO16
9	MTMS	I/O	GPIO 14; HSPI_CLK
10	MTDI	I/O	GPIO 12; HSPI_MISO
11	VDDPST	P	Digital/IO Power Supply (1.8V ~ 3.6V)
12	MTCK	I/O	GPIO 13; HSPI_MOSI; UART0_CTS

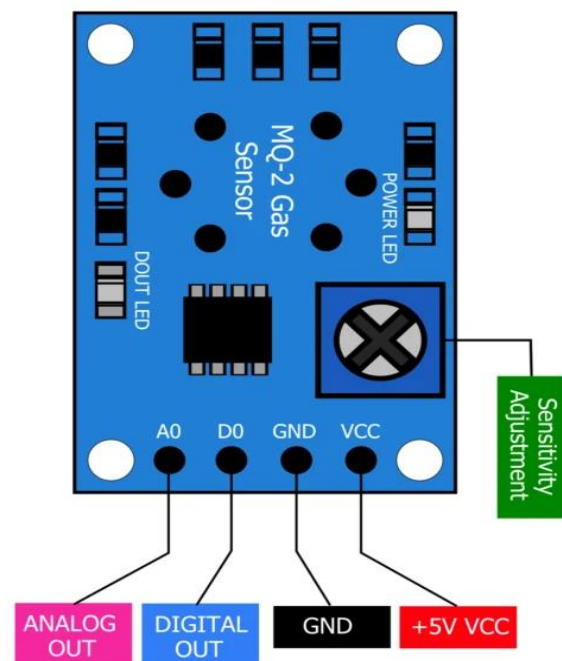
13	MTDO	I/O	GPIO 15; HSPI_CS; UART0_RTS
14	GPIO2	I/O	UART TX during flash programming; GPIO2
15	GPIO0	I/O	GPIO0; SPI_CS2
16	GPIO4	I/O	GPIO4
17	VDDPST	P	Digital/IO Power Supply (1.8V ~ 3.6V)
18	SDIO_DAT A_2	I/O	Connect to SD_D2 (Series R: 200Ω); SPIHD; HSPIHD; GPIO9
19	SDIO_DAT A_3	I/O	Connect to SD_D3 (Series R: 200Ω); SPIWP; HSPIWP; GPIO10
20	SDIO_CMD	I/O	Connect to SD_CMD (Series R: 200Ω); SPI_CS0; GPIO11
21	SDIO_CLK	I/O	Connect to SD_CLK (Series R: 200Ω); SPI_CLK; GPIO6
22	SDIO_DAT A_0	I/O	Connect to SD_D0 (Series R: 200Ω); SPI_MISO; GPIO7
23	SDIO_DAT A_1	I/O	Connect to SD_D1 (Series R: 200Ω); SPI_MOSI; GPIO8
24	GPIO5	I/O	GPIO5
25	U0RXD	I/O	UART Rx during flash programming; GPIO3

26	U0TXD	I/O	UART TX during flash programming; GPIO1; SPI_CS1
27	XTAL_OUT	I/O	Connect to crystal oscillator output, can be used to provide BT clock input
28	XTAL_IN	I/O	Connect to crystal oscillator input
29	VDDD	P	Analog Power 2.5V ~ 3.6V
30	VDDA	P	Analog Power 2.5V ~ 3.6V
31	RES12K	I	Serial connection with a 12 k Ω resistor and connect to the ground
32	EXT_RSTB	I	External reset signal (Low voltage level: active)

Front View



Back View



PIN DESCRIPTION OF GAS SENSOR

MQ2 Sensor Pin Configuration

Pin No:	Pin Name:	Description
For Module		
1	Vcc	This pin powers the module, typically the operating voltage is +5V
2	Ground	Used to connect the module to system ground
3	Digital Out	You can also use this sensor to get digital output from this pin, by setting a threshold value using the potentiometer
4	Analog Out	This pin outputs 0-5V analog voltage based on the intensity of the gas
For Sensor		
1	H -Pins	Out of the two H pins, one pin is connected to supply and the other to ground
2	A-Pins	The A pins and B pins are interchangeable. These pins will be tied to the Supply voltage.
3	B-Pins	The A pins and B pins are interchangeable. One pin will act as output while the other will be pulled to ground.

PIN DESCRIPTION OF 1 CHANNEL RELAY